Thermal extremes and cardiovascular first-hospitalisation burden in Hong Kong, 2013–2023

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Exploratory monthly ecological panel ($N = 132$) · HA_APPROVED_AGGREGATE outcomes · **Gate 3 OPEN — exploratory signals, no confirmatory claim** · ISO A0 portrait (841 mm × 1189 mm)

1. Why this question

Hong Kong is heating fastest at night, yet winter cold snaps persist. Overnight heat impairs physiological recovery; cold stress raises blood pressure, viscosity, and autonomic load. For a high-risk cardiometabolic population, the burden that reaches hospital—not only mortality—is what adaptation planning must budget for.

- Hot nights at HKO Headquarters rose from 10 (2013) to a 61-day peak (2021)
- Cold days did not disappear: 13–14 per year even in 2021–2023
- The 65–74 population nearly doubled over the same decade

The lab's mortality work (Liu 2020; Liu 2026) frames the *mortality* layer; this project asks the complementary *morbidity* question.

Question. Are monthly thermal-burden metrics associated with the governed count of *first* CHD or heart-failure hospitalisations after first diagnosis in a T2D/HTN cohort?

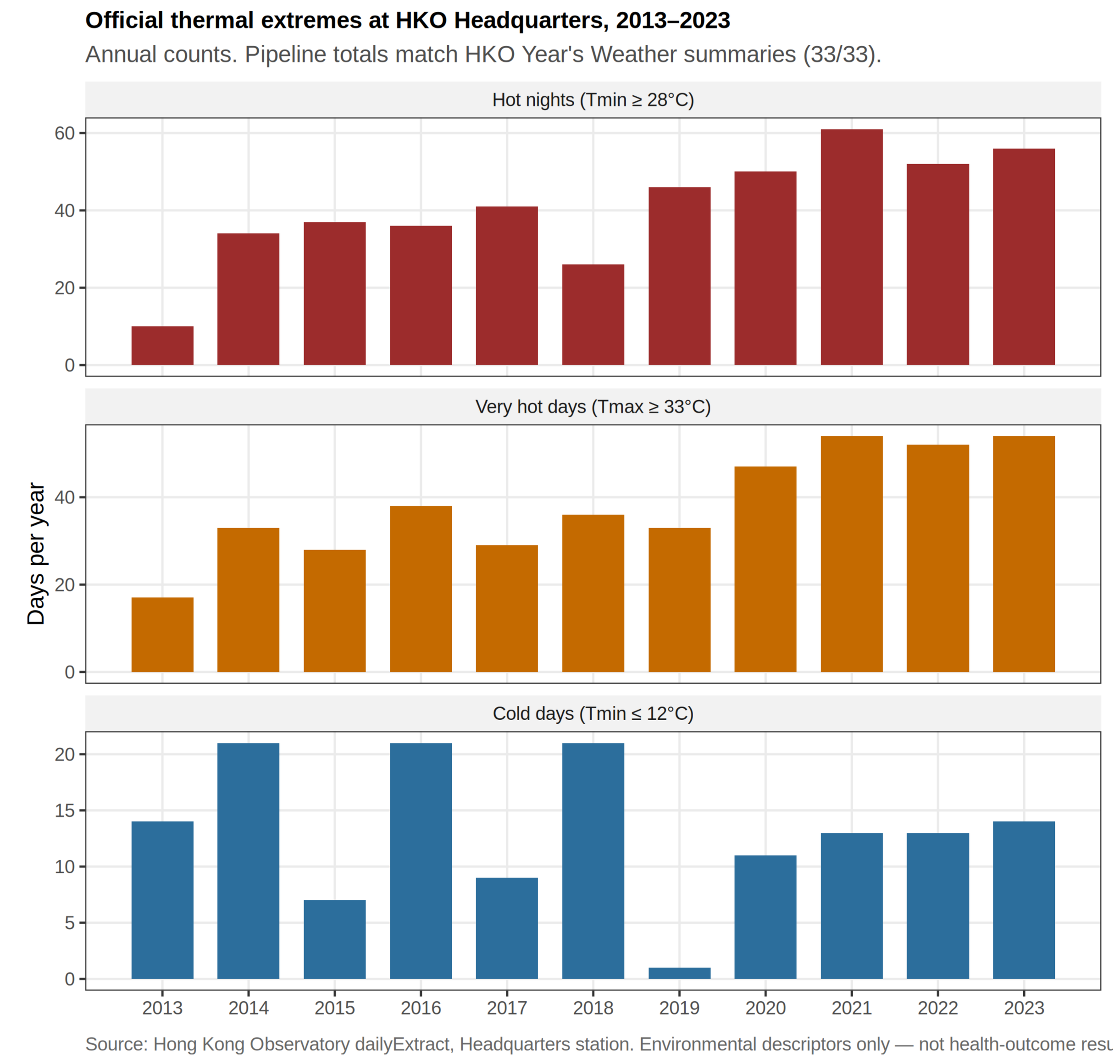
Scope boundaries. This is **not** an AMI / principal-diagnosis study (admission cause not recorded). The stroke file was **not delivered** — no stroke result is claimed anywhere on this poster.

2. Data, cohort & estimand

Domain	Source	Provenance
Weather	HKO Headquarters daily; official extremes	REAL (33/33 match)
Pollution	EPD NO_2, O_3, particulates	REAL
Population	C&SD mid-year estimates	REAL
Outcomes	HA first CHD/HF hospitalisation after first diagnosis, T2D/HTN cohort	HA_APPROVED_AGGREGATE
Software	R; scripted pathways 31–34; machine-validated release	Reproducible

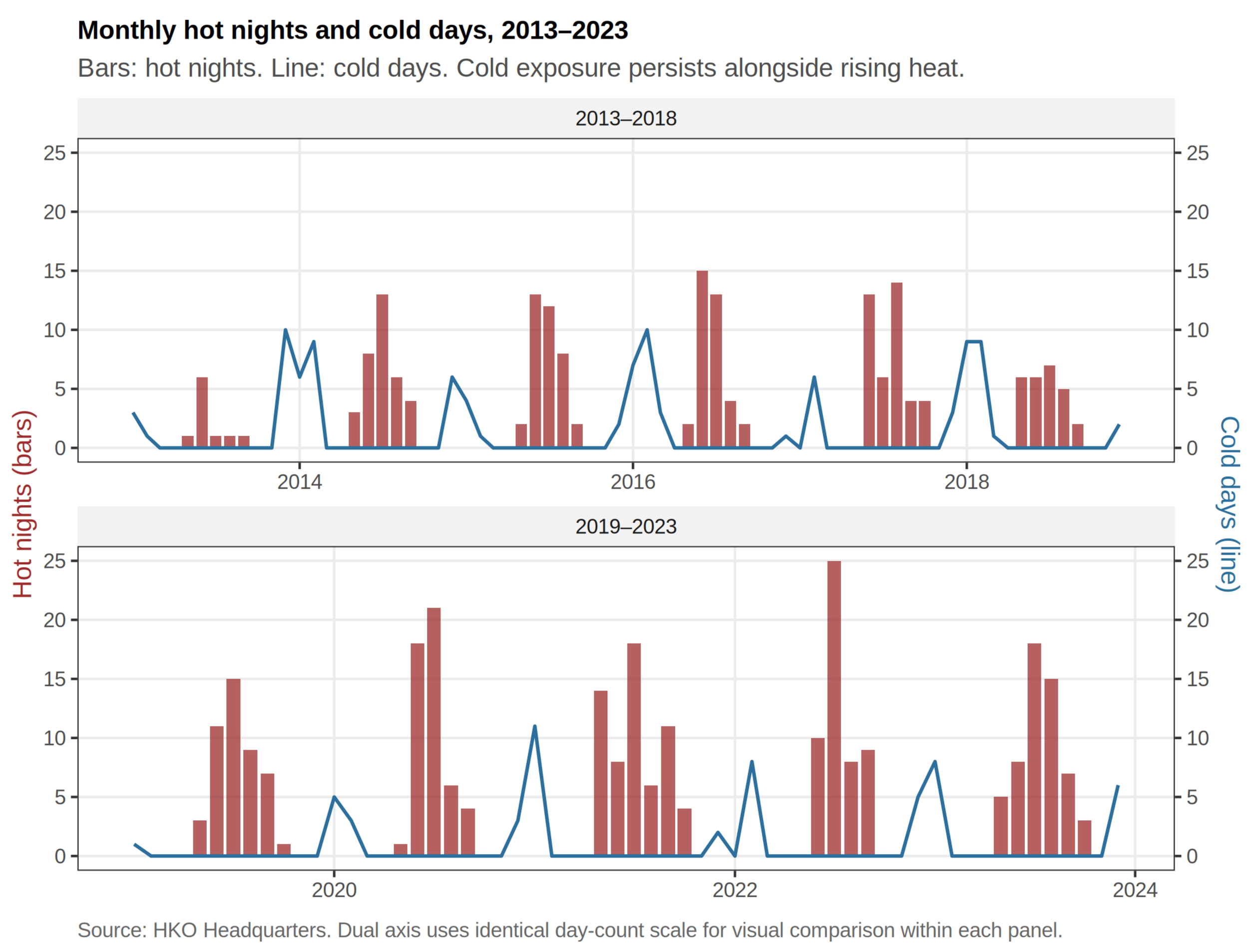
- **Window:** 132 territory-months, Jan 2013 – Dec 2023
- **Events:** 156,156 CHD; 29,681 HF first hospitalisations
- **Estimand:** monthly count ratio per exposure increment — per 5 days for extremes, per 1°C for continuous metrics; days-in-month offset
- **Model:** negative binomial; calendar-month indicators + trend spline

3. Heat rose; cold did not disappear (REAL)



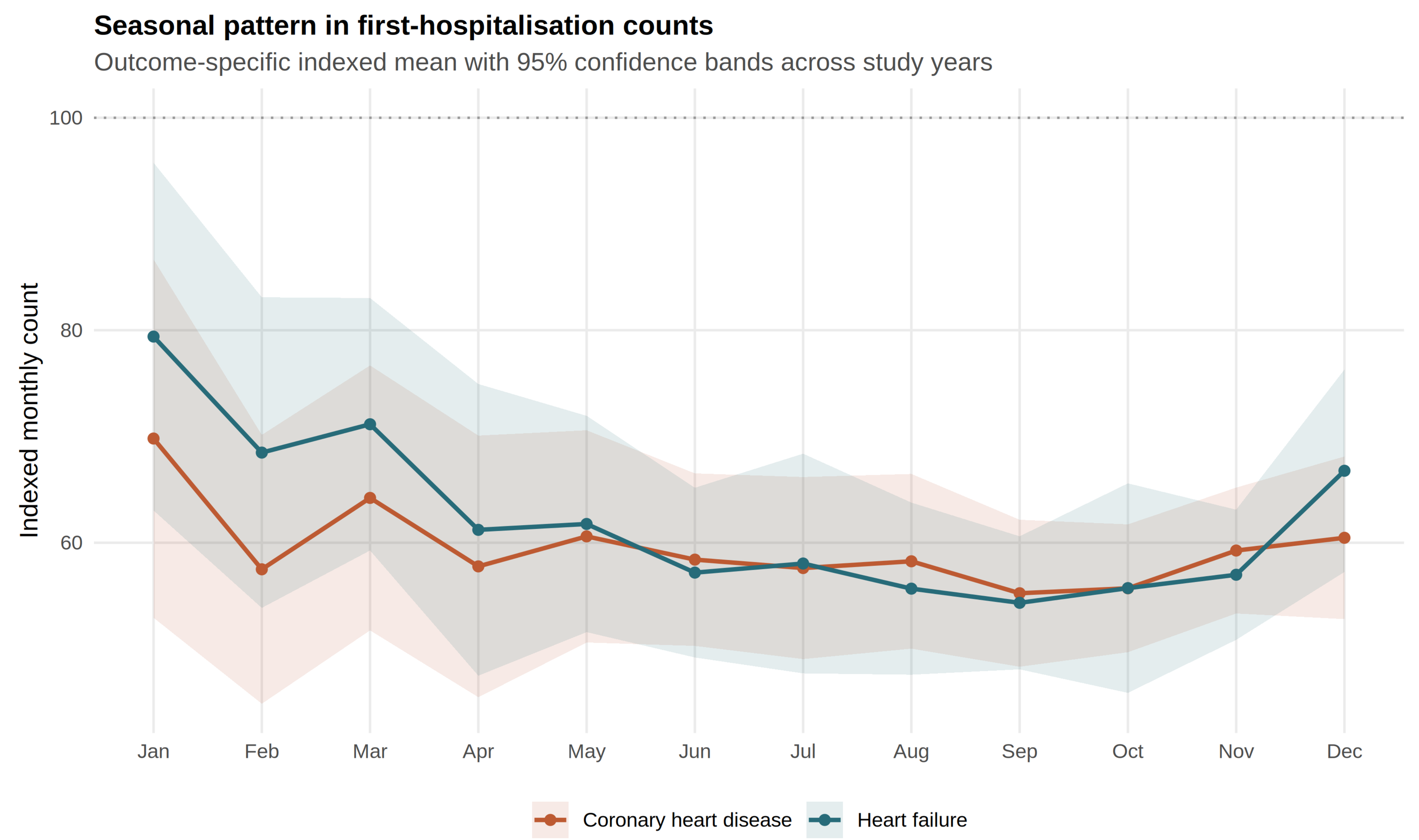
Annual counts of official HKO extremes at Headquarters. Environmental descriptors only — **not** health-outcome results. Pipeline totals match HKO *Year's Weather* summaries 33/33.

4. Monthly hot nights and cold days (REAL)



Monthly official extreme-day counts, 2013–2023: hot nights (bars) climb while cold days (line) recur every winter. Weather descriptors only.

5. Seasonal pattern of first hospitalisations



Outcome-specific indexed monthly mean with 95% bands (HA_APPROVED_AGGREGATE; indexed display — raw monthly counts are not shown). Both outcomes peak in winter and trough in early autumn; HF seasonality is the stronger of the two.

6. Literature map — estimands are not interchangeable

Study	Quantity	Scale
Liu (J.) 2020, *SCS*	CVD **mortality** attributable fraction; cold fraction > heat fraction	Daily / annual AF
Liu (Z.) 2026, medRxiv	Modelled heat-related **excess deaths**	Daily mortality
Goggins 2012/2013	Cold–stroke / cold–AMI, % per °C cooling	Daily, lagged
Guo 2024	Binary hot-night flag null for excess hospitalisation; night-heat *intensity* informative	Daily
This project	**First CHD/HF hospitalisation**, count ratio per 5 extreme days	**Monthly ecological**

Prior Hong Kong work quantifies *mortality* fractions and *daily* triggering. This project adds the governed *monthly morbidity* layer for first CHD/HF hospitalisations — a neighbouring estimand, not a replication.

A daily per-°C triggering coefficient, a mortality attributable fraction, and a monthly morbidity count ratio answer different questions. None may be substituted for another.

7. Methods — one 12-contrast panel

- **Panel:** 2 outcomes (CHD, HF) × 6 exposures (mean, Tmax, Tmin per 1°C; hot nights, cold days, very hot days per 5 days) = 12 pre-specified contrasts
- **Exposures** (official HKO definitions): hot night Tmin ≥ 28°C; very hot day Tmax ≥ 33°C; cold day Tmin ≤ 12°C
- **SE ladder** per contrast: Model → HC1 → NW3 → NW6; continuity decisions use **NW6**, full ladder shown
- **Multiplicity:** Benjamini–Hochberg q across the 12 core contrasts
- Reported as one panel — not twelve independent "discoveries"
- Full pipeline scripted and re-runnable; release package machine-validated before any number reached this poster

8. Key exploratory results (Gate 3 open)

1.022 (1.002–1.042)
CHD × hot nights per 5 days; NW6 95% CI; $p \approx 0.032$; $q = 0.192$

1.073 (1.006–1.144)
HF × cold days per 5 days; NW6 95% CI; $p \approx 0.031$; $q = 0.192$

- All 12 core Benjamini–Hochberg q-values > 0.19 — **no contrast survives multiplicity control**
- The CHD hot-night interval *includes the null* under Model and HC1 standard errors — the signal is **SE-sensitive**
- Provisional direction pattern (CHD–heat nights; HF–cold) is flagged for the Gate 3 discussion, not promoted to a finding

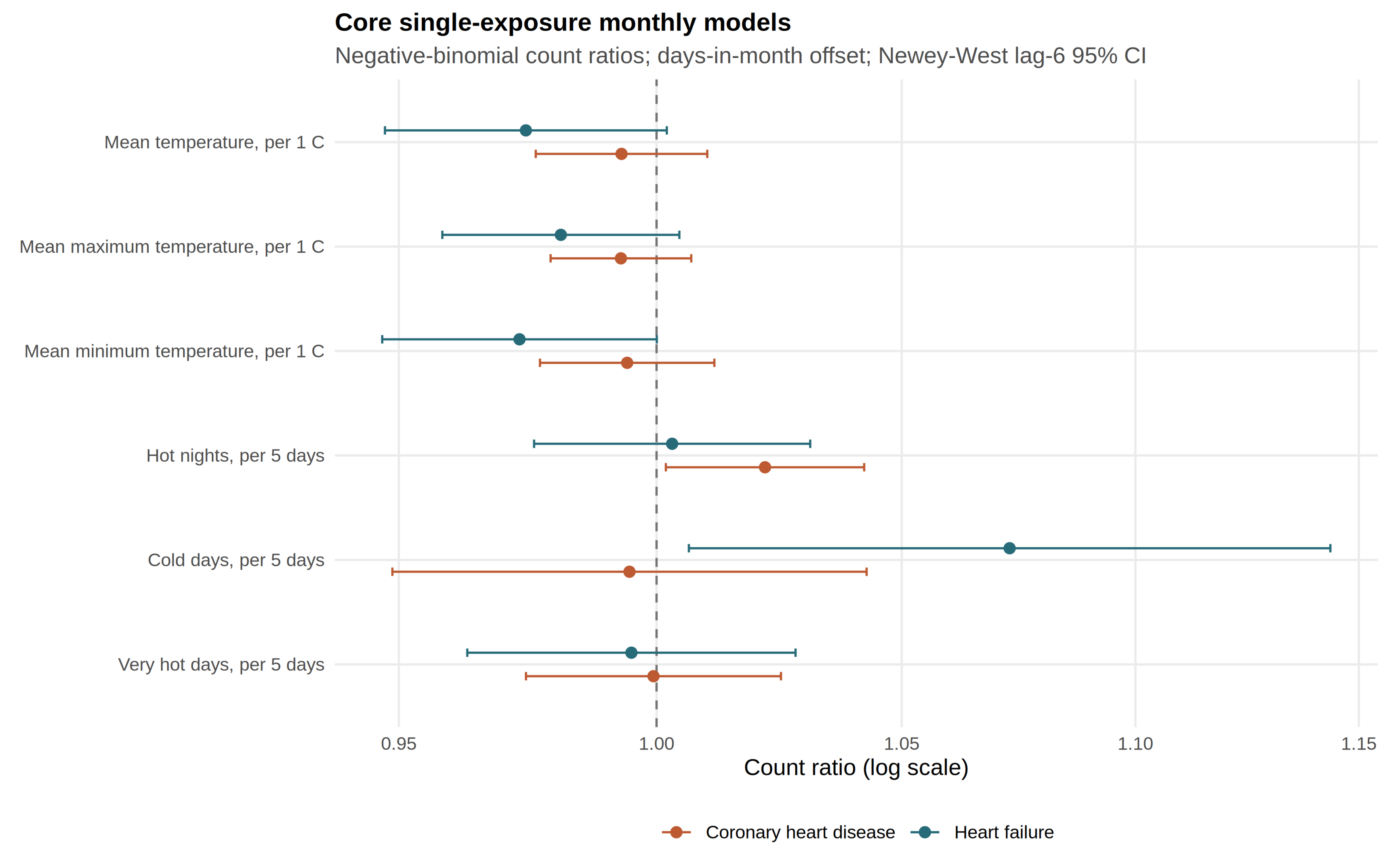
These are exploratory signals. Gate 3 is open; no confirmatory headline is claimed on this poster or elsewhere.

9. The 12-contrast core at a glance (NW6, exploratory)

Outcome	Exposure	Count ratio (95% CI)
CHD	Hot nights / 5 days	**1.022 (1.002–1.042)**, $q = 0.192$
CHD	Cold days / 5 days	0.995 (0.949–1.043), near null
CHD	Very hot days / 5 days	0.999 (0.974–1.025), near null
CHD	Mean / Tmax / Tmin per °C	≈0.993–0.994, near null
HF	Cold days / 5 days	**1.073 (1.006–1.144)**, $q = 0.192$
HF	Hot nights / 5 days	1.003 (0.976–1.031), near null
HF	Mean Tmin / °C	0.973 (0.947–1.000), borderline
HF	Very hot days / 5 days	0.995 (0.963–1.028), near null

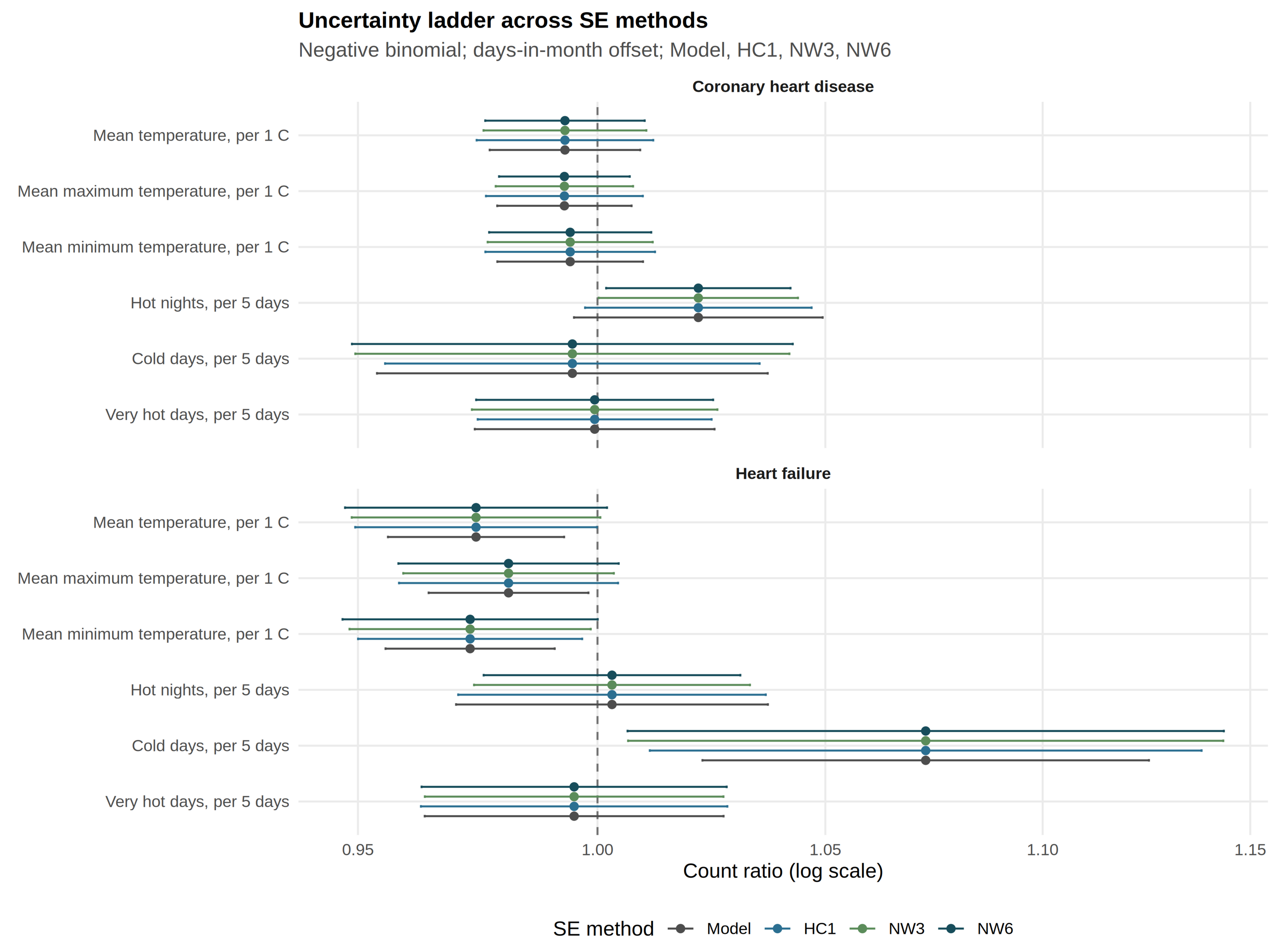
Amended single-exposure core; days-only offset; NW lag-6. Values as validated in the release panel; do not quote as primary without the team freeze.

10. Core single-exposure models (exploratory)



Negative-binomial count ratios, days-in-month offset, Newey–West lag-6 95% intervals. HA_APPROVED_AGGREGATE outcomes; Gate 3 open — exploratory.

11. Uncertainty ladder across SE methods



The same 12 contrasts under Model, HC1, NW3 and NW6 standard errors. The HF–cold signal persists across the ladder; the CHD hot-night interval crosses the null under Model/HC1 — reported rather than hidden.

12. Feasibility note — a negative methods result

A constrained monthly→daily (M | D) downscaling calibration was run under strict gates (F1.2). It **failed**: no real daily coefficient was admitted. This bounds what the method can deliver from monthly aggregates — a methods result, **not** a health finding. The monthly panel remains the resolution of record.

13. Limitations, stated plainly

- Ecological territory-month design — no individual-level inference
- HKO Headquarters is a single-station proxy for territory-wide exposure
- Monthly resolution captures *burden*, not day-level triggering
- Admission cause unrecorded: outcomes are first hospitalisation *after* first CHD/HF diagnosis, not cause-specific events
- Multiplicity control limits per-contrast power; near-null intervals do not exclude small effects
- All estimates provisional until the Gate 3 team freeze

14. Where to read more

Manuscript	manuscript/chd_hf_thermal_associations_2013_2023.md
Gate 3 packet	reports/gate3_decision_packet_2026-08-07.md
Results panel	reports/laidlaw_stage3/results_panel_chd_hf_2026-08-07.md
Release	outputs/release_chd_hf/ (figures, tables, diagnostics)
Integrated report	reports/bishai_integrated_report/

Provenance labels used throughout: **REAL** (HKO, EPD, C&SD), **HA_APPROVED_AGGREGATE** (HA outcomes), **SYNTHETIC** (dev-only; quarantined, never reported as findings).

15. Take-home

1. A reproducible, provenance-labelled monthly panel for CHD/HF first-hospitalisation burden now exists end-to-end (132 months; REAL weather / population; HA_APPROVED_AGGREGATE outcomes; release package re-validated).
2. Two exploratory signals — CHD with hot nights, HF with cold days — are hypothesis-generating only while Gate 3 remains open.
3. Honest refusals: no stroke (file not delivered), no principal-diagnosis AMI (cause not recorded), no protected primary claim before the team freeze.
4. Next executable step: the team Gate 3 review of the headline candidates; stroke analysis begins only if and when the governed file arrives.

What closes Gate 3: a team decision freezing (or dropping) the headline contrast set, with the SE ladder and diagnostics reviewed together.

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